FUNDAMENTAL AND DERIVED QUANTITIES, STANDARD PREFIXES, UNIT CONVERSIONS

In this chapter basic physical quantities used in the study of Physics are discussed. Standard units for the quantities mentioned are also given. Also, ways of writing very small or very large numbers are discussed. Writing such numbers in standard form helps you to avoid mistakes. Employing standard prefixes also simplifies your calculations. Measurement of length, time, temperature, mass and weight will be discussed. Instruments used for measuring each of these quantities are specified. And also dimension analysis is carried out to check the correctness of some of the equations used for calculations in Physics.

Objectives

At the end of this session you should be able to:

1. explain what physical quantities are and give examples of each with their standard units
2. explain what Fundamental and derived quantities are and give examples of each with their units;
3. memorize some of the standard prefixes:
4. write very large and very small numbers in standard forms using the standard prefixes;
5. to convert one unit scale to another.
6. Identify the correct instrument required to measure any of the fundamental quantities;
7. Explain how the various fundamental quantities are measured;
8. Demonstrate these measurements practically.
9. Use dimensional analysis to verify the correctness of equations and units.

1.1 Physical Quantities and Units

Physics is sometimes defined as the study of energy and matter and their relationships. Physics is a natural science that also deals with measurement of physical quantities. We define a physical quantity either by specifying how it is measured or by stating how it is calculated from other measurements. For example, we define distance and time by specifying methods for measuring them, whereas we define average speed by stating that it is calculated as the ratio of distance traveled to the time taken.

Measurements of physical quantities are expressed in terms of *units*, which are standardized values. For example, the length of a ladder, which is a physical quantity, can be expressed in units of meters. Without standardized units, it would be extremely difficult for scientists to express and compare measured values in a meaningful way. Thus, any number or set of numbers used for the quantitative description of a physical phenomenon is called a physical quantity.

1.2 Fundamental Quantities and Units

The fundamental units are the units of the fundamental quantities, as defined by the International System of Units. They are not dependent upon any other units, and all other units are derived from them. In the International System of Units, the fundamental units are:

Table 1.1 SI units for fundamental quantities

Quantity	Unit Name	Unit Symbol
Length	Meter	m
Time	Second	s
Mass	Kilogram	kg
Electric current	Ampere	A
Amount of substance	Mole	mol
Light intensity	Candela	Cd
Temperature	Kelvin	K

1.3 Derived Quantities and Units

Derived quantities are those that may be expressed in terms of base or derived quantities by means of the mathematical symbols of multiplication and division only (no addition or subtraction or any other sign). Basically they may be divided into two categories. The first group consists of derived quantities, which have proper units. Their units can be expressed in terms of units of base or derived units. The second class consists of quantities that do not have units expressed in terms of base or derived units.

Below are examples of SI derived units in terms of base units;

Derived Unit	Symbol	Name of Unit	Symbol of Unit
Area	A	Square meter	m^2
Volume	V	Cubic meter	m^3
Speed/velocity	v	metre per second	m/s
Acceleration	a	metre per second squared	m/s^2
Specific volume	ν	Cubic meter per kilogram	m^3/kg
Current density	j	ampere per meter squared	A/m^2
Amount concentration	c	mole per cubic meter	mol/m^3
Density	ρ	kilogram per meter cube	Kg/m^3
Luminance	L_v	candela per meter square	cd/m^2

Standard Prefixes

In physics, we sometimes deal with numbers that are very small as well as numbers that are very large. It is not convenient to write very small or large numbers in the conventional decimal notation. Once the fundamental units are defined, it now becomes easy to introduce larger and smaller units for the same physical quantities. In the metric system measurements additional units always relate to the fundamental units by multiples of 10 or $\frac{1}{10}$. The multiplicative factors are most conveniently expressed in exponential notation; thus $10000 = 10^4, \frac{1}{100} = 10^{-2}$. The names of the additional units are always derived by adding a prefix to the name of the fundamental units.

Table 1.2 Prefixes for the power of ten

Power of ten	Prefix	Abbreviation
10^{-24}	yocto-	Y
10^{-21}	zepto-	Z
10^{-18}	atto-	A
10^{-15}	fermto-	F
10^{-12}	pico-	P
10^{-9}	nano-	N
10^{-6}	micro-	M
10^{-3}	milli-	M
10^{-2}	centi-	C
10^{3}	kilo-	K
10^{6}	mega-	M
10^{9}	giga-	G
10^{12}	tera-	T
10^{15}	peta-	P
10^{18}	exa-	E
10^{21}	zetta-	Z
10^{24}	yotta-	Y

The following are a few examples of the use of the multiples of ten and their prefixes.
1 nanometer = 1nm = 10^{-9} m
1 micrometer = 1μm = 10^{-6} m
1 millimeter = 1mm = 10^{-3} m
1 centimeter = 1cm = 10^{-2} m
1 kilometer = 1km = 10^{3} m
1 milligram = 1mg = 10^{-6}kg
1 gram = 1g = 10^{-3} kg
1 nanosecond = 1ns = 10^{-9} s
1 microsecond = 1ms =10^{-3} s

Unit Conversions
Relations among physical quantities are often expressed by equations in which physical quantities are represented by algebraic symbols. An algebraic symbol always denotes both a number and a unit. For example, *x* might represent distance of 5m, *t* a time of 30s and *m* a mass of 10kg. We often need to change the units in which a physical quantity is expressed. We do so by multiplying the original measurement by a conversion factor which is a ratio of the units in question that equals unity. For example

$$\frac{1\text{min}}{60s}=1 \text{ and } \frac{60s}{1\text{min}}=1$$

This is so because multiplying any physical quantity (or algebraic symbol) by unity leaves the quantity unchanged. For example we can convert 5min into seconds by multiplying it by the conversion factor of $\frac{60s}{1\min}$ to have

$$5\min = (5\min)(1) = (5\min)(\frac{60s}{1\min}) = 300s$$

MEASUREMENT OF FUNDAMENTAL QUANTITIES, DIMENSIONS AND UNITS ANALYSIS

1.4 Measurement of Length

Length is defined as the straight-line distance between two points along an object or in space. The SI unit of length is the meter. Different types of lengths are measured by using different types of instruments. Lengths like the length of cloth or length of a line can be measured by using measuring tape, a metre scale or a foot rule. But these instruments cannot be used to measure the diameter of a metal sphere or a cylinder. Micrometer screw gauge and Vernier calipers are the appropriate instruments used to measure the diameter of cylindrical objects.

1.5 Measurement of Time

Time is defined as the period in which a process, action or an event takes place. *Time* is also called the Point or Period when something occurs. Time is a scalar quantity and its SI unit is the second (s). The instruments used for measuring time are stop clocks and stop watches.

1.6 Measurement of Temperature

Temperature is a measure of the average kinetic energy of the particles of a substance. The higher the temperature of an object, the higher is its kinetic energy. Kinetic energy is a type of energy associated with motion. Temperature is measured with thermometers, which are devices that use the expansion of a substance to give an indirect measure of temperature. The units of temperature are called degrees Celsius (^oC), degrees Fahrenheit (^oF) or Kelvin (K).

1.7 Measurement of Mass and Weight

Mass is a measurement of the quantity of matter contained in a body. The mass of an object is measured in kilogram (kg). Mass of objects can be measured by using a beam balance or an electronic balance.

The weight of an object is proportional to the gravitational force of the earth on the object. Therefore Weight is the measurement of the pull of gravity on an object. Weight of an object can be determined with a spring balance. Weight of a body is directly proportional to it Mass. The more the mass of an object, the more weight it has.

1.8 Dimensions and Unit Analysis

The fundamental quantities used in physical descriptions are called dimensions. For example length, time and mass are dimensions. You could measure the distance between two points and express it in units of meters, centimeters, or feet, but the quantity would still have the dimension

of length. It is common to express dimensional quantities by bracketed symbols as [L], [M] and [T] for length, mass and time respectively. Derived quantities are a combination of dimensions. For example, velocity v has dimensions of $[L]/[T]$ (think of it as meters/second = length/time) and volume, V has dimensions $[L]\times[L]\times[L]$, or $[L^3]$. Any equation must always be dimensionally consistent or homogenous-this means that any two terms added or subtracted in an equation must have the same units. For example given the equation: 3.0m + 2.0m = 5.0m (similar to $a+b=c$), comparing both sides we have $[L]+[L]=[L]$ or in words length plus length gives you another length not 2[*L*]. The above procedure for checking units in equations is called dimensional analysis. It is formally defined as a procedure by which the dimensional consistency of any equation may be checked. One major advantage of dimensional analysis is its usefulness for checking whether an equation that has been derived or being used to solve a problem has the correct form. The dimensions and units of some common quantities are given in Table 1.3.

Table 1.3 Dimensions and Units

Quantity	Dimension	Unit
Mass	$[M]$	kg
Time	$[T]$	s
length	$[L]$	m
Area	$[L^2]$	m^2
volume	$[L^3]$	m^3
velocity	$[L]/[T]$	m/s
Acceleration (a or g)	$[L]/[T^2]$	m/s^2

Example 1.1. Check if the following equations are dimensionally correct:

(a) $x=at$ (b) $x=at^2$

Solution:

(a) Equation $x=at$ can be expressed dimensionally as

$[L]=\frac{[L]}{[T^2]}\times[T]=\frac{[L]}{[T]}$ which is not true. Thus $x=at$ cannot be a correct equation.

(b) Equation $x=at^2$ can be expressed dimensionally as

$[L]=\frac{[L]}{[T^2]}\times[T^2]=[L]$ this equation is dimensionally correct but as you will see later it is not physically correct. The correct form of $x=at^2$ both dimensionally and physically is $x=\frac{1}{2}at^2$. The fraction $\frac{1}{2}$ has no dimensions i.e. it is a dimensionless number. Doing dimensional analysis with the symbols is fine but in practice it is often more convenient to use actual unit abbreviations such as m, s, and kg. Units can also be treated as symbols and cancelled. Using units instead of symbols in dimensional analysis is called units analysis.